

LUCAS DELBECQUE

COMPUTER SCIENCE STUDENT - AI - DATA SCIENCE
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Portfolio link : https://lucasdelbecque.github.io/portfolio/

Rigorous and creative Computer Science student with a strong background in AI, Software Engineering, and 3D Development. Seeking a final-year internship starting September 2026 to apply my technical expertise to innovative projects.

FORMATION

Lycée Chaptal, Paris - Scientific preparatory class *2021 - 2022*

UTC, Compiègne - Computer Engineering degree, Specialization in artificial intelligence and data science *2022 - 2027*

- Machine learning, Deep learning, Feature engineering, Multi-agents systems, Cloud and Network, Web Programming, Object-Oriented Programming, Project management

UFSC, Florianopolis - Computer science Master, university exchange *2026*

WORK EXPERIENCE

Data Scientist intern - Shift - Paris

- Developped new fraud detection tools for insurance clients
- Optimized LLM prompts to detect subrogation cases in insurance claims

Software Engineer intern - SoWhen? - Montreuil

- Developed interactive web applications and Augmented Reality experiences.
- Developed a hand gestures recognition system in Augmented Reality with Unreal Engine
- Collaborated on the full software development lifecycle to deliver immersive digital solutions.

Web developer - Nossa

- Designed and developed a showcase website (vitrine) to enhance digital presence and brand identity.
- Optimized user interface (UI) and performance to drive online engagement.

Maintenance Technician, Video editor -Blue Spirit, Paris

- Performed technical maintenance and assembly of specialized equipment.
- Ensured operational efficiency and followed strict safety and quality standards.

ASSOCIATIVE ENGAGEMENT

Pic'Asso, Student association - Compiegne, France

- IT Student Team: Managed a custom-built POS platform, self-service payment kiosks, and media distribution systems.

SKILLS

- Programming langages : C++, C, Python, Java, SQL, Go
- Framework & librairies : Scikit-learn, TensorFlow, PyTorch, Qt, SpringBoot, OpenGL, Metal
- Web Developement : HTML, CSS, JavaScript, React, JavaFX
- Data & deployment tool : Docker, Rasperry Pi
- AI & Machine Learning: Hugging Face, RAG, CNN, Supervised/Unsupervised Learning
- Software & Creative Tools: Unreal Engine, DaVinci Resolve, Ray Tracing, Path tracing

PROJECTS

- **GPU Rendering Engine :** Developed a mini-rendering engine (C++, OpenGL, GLSL) featuring spatial convolution filters.
- **Interactive OCR Tool :** Created an interactive OCR tool based on a Convolutional Neural Network (CNN), combining Python training with a C++/OpenGL inference engine deployment.
- **Blackhole real time simulation:** Blackhole simulation using GPU Ray tracing and Schwarzschild equations.
- **GPU Monte Carlo Path Tracer :** Implemented a physically-based GPU path tracer using Metal compute shaders.
- **CPU Ray Tracer from Scratch :** Built a multithreaded physically-based path tracer in pure C++.
- **Real-Time PBR Showcase :** Real-time 3D renderer built with OpenGL/GLSL implementing a full deferred rendering pipeline
- GPU Fluid Simulation : SPH fluid simulation using GPU via Metal compute shaders.
- **Studio-Brain AI Assistant (RAG):** Developed a local Generative AI solution using LangChain and Ollama.
- **Multi-agent Simulation (Go):** Simulation of a shopping mall.
- **Video Game & Cinematic Creation:** Created video games and cinematics using Unreal Engine.
- **Film Production:** Writing, directing, and editing short films.

LANGUAGES

French : Native
English : C1 TOEFL
Portuguese : B1
German : A2

HOBBIES

Basketball
Cinema
3d production
Creative Arts